

AUTORAG: THE SELF-HEALING KNOWLEDGE ENGINE

AWS IMPACTX CHALLENGE | IIT BOMBAY HACKATHON

TEAM NAME: AUTORAG INNOVATORS

TEAM: NEHA SHARMA, AMISHA LALWANI, KACHIGALLA
VAISHNAVI, SRAVANTHI VADLA



THE PROBLEM

Modern RAG systems break silently:

- Hallucinate when information is missing or outdated
- Require engineers to manually fetch, clean, and re-index data
- Not suitable for dynamic, fast-changing environments

Who suffers?

- Universities (helpdesk bots)
- HR/Policy Teams
- Developer Documentation & Support
- Enterprise Knowledge Portals



THE SOLUTION

A fully autonomous RAG engine that patches its knowledge base in real-time.

Key Capabilities:

- Detects low-trust answers automatically
- Launches external search + crawling
- Filters, embeds, and inserts new knowledge instantly
- Re-answers using enriched index
- Continuous reinforcement improves retrieval

TECHNICAL APPROACH

1. Query Processor converts user query → normalized search request
2. Retriever fetches top-k vectors
3. Trust Score Evaluator computes similarity, context quality & confidence
4. If low trust → Self-Healing Controller triggers:
 - DuckDuckGo/Google search
 - PDF/HTML crawling
 - Relevance ranking via LLM
5. Chunk + embed + live upsert into ChromaDB
6. LLM generates improved answer
7. Feedback engine scores useful docs → improves future ranking

TECH STACK & SYSTEM ARCHITECTURE

Frontend:

- Streamlit, React (optional)

Backend & Orchestration:

- FastAPI, Python, LangChain, LangGraph

Embeddings & Vector DB:

- MiniLM-L6-v2, BGE-small-en
- ChromaDB

LLMs:

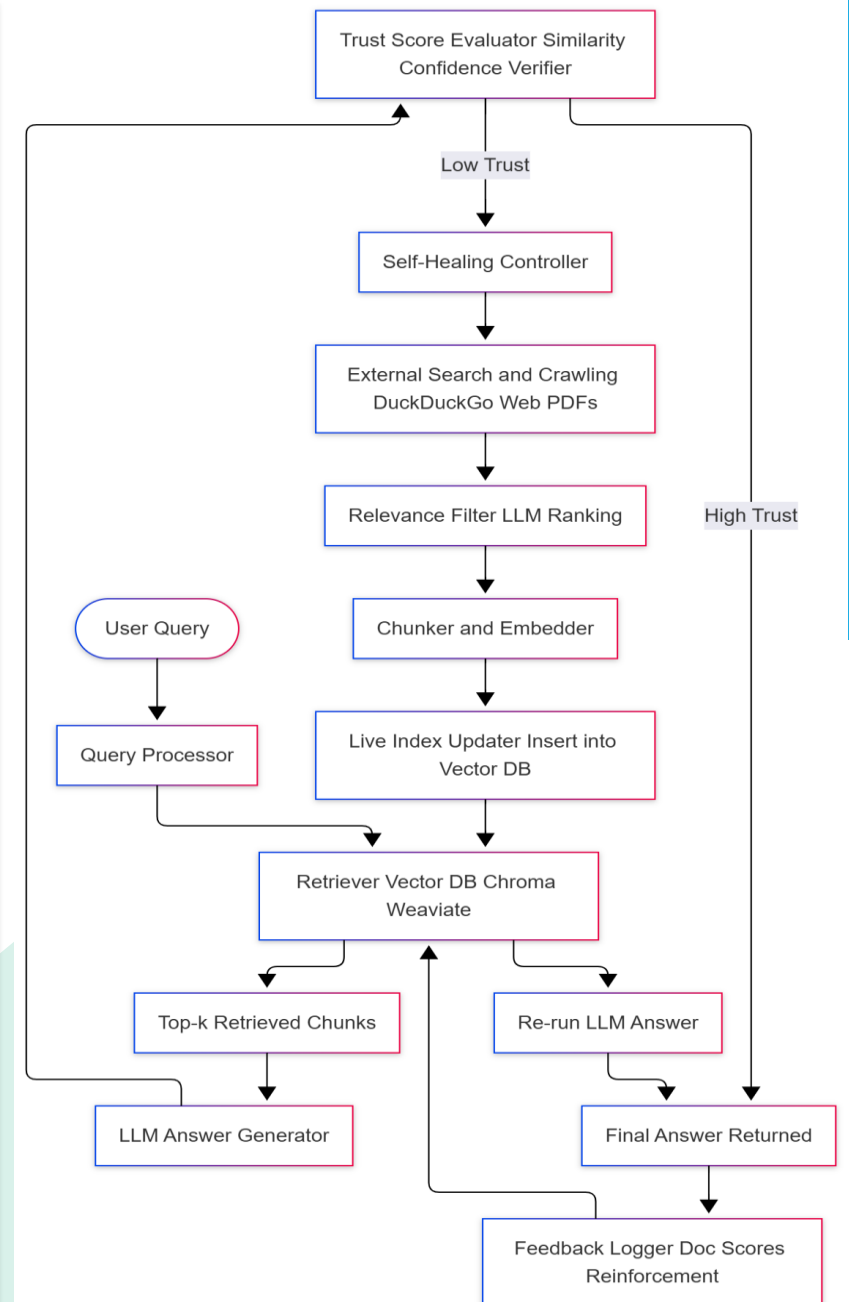
- Groq Llama 3.1 (8B/70B)
- GPT-4o-mini

Retrieval:

- DuckDuckGo Search API
- Serper.dev
- BeautifulSoup, PDFplumber

Feedback & Monitoring:

- BM25/BGE Re-ranker
- SQLite/JSON logs
- Stream-lit dashboard



FEASIBILITY & VIABILITY

Technical Feasibility:

- Uses lightweight open-source tech (Chroma, MiniLM) → low compute
- Modular pipeline deployable on AWS Lambda / EC2
- Groq LLM inference → ultra-fast answers

Operational Viability:

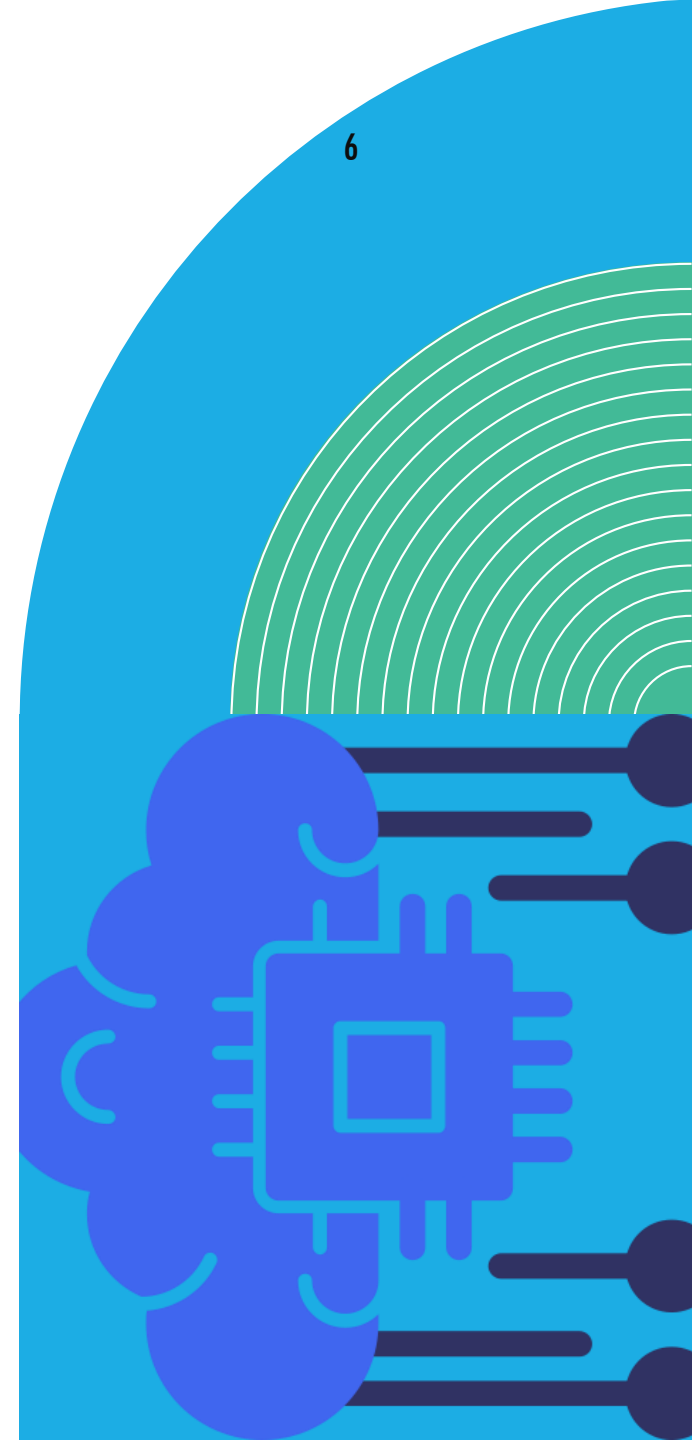
- Auto-updating KB reduces maintenance by ~80%
- Plug-and-play for university, enterprise or public portals

Business Viability:

- SaaS deployment possible
- Can scale to multi-tenant architectures

IMPACTS & BENIFITS

- Eliminates hallucinations caused by stale indexes
- Makes RAG self-sustaining & autonomous
- Delivers reliable, traceable answers → boosts user trust
- Reduces cost of manual data ingestion
- Ideal for helpdesks, student portals, HR systems, govt knowledge bases



RESEARCH & REFERENCES

- Stanford RAG Survey (2024) – dynamic RAG limitations
- Meta Llama 3.1 Research
- Reimers & Gurevych (2019): Sentence Transformers
- ChromaDB performance documentation
- DuckDuckGo Search API specs
- LangChain indexing & retrieval best practices

